

HYDROFARMS AGRI SOLUTIONS

MASTER CONSTRUCTION METHOD STATEMENT

One-Feddan NFT Lettuce Greenhouse Project

Document field	Controlled value
Document code	HF-GH01-MST-GEN-001
Revision	00 - Draft for coordination
Project	NFT lettuce production greenhouse and service greenhouse
Location	Alexandria, Egypt
Prepared by	Hydrofarms Agri Solutions
Issue date	15 August 2026
Language	English

CONTROLLED DOCUMENT
This master document defines the project-wide construction methodology and control system. Activity-specific method statements, approved-for-construction drawings, supplier instructions, risk assessments and inspection records remain mandatory and take precedence where they provide greater detail.

Document Control

Approval and distribution

Role	Name / organisation	Action
Prepared by	Hydrofarms project engineering	Technical preparation and coordination
Reviewed by	Project manager / discipline leads	Constructability, interfaces and resources
HSE review	HSE manager	Risk controls and permit requirements
QA/QC review	QA/QC manager	Inspection, testing and records
Approved by	Client / consultant authorised representative	Approval prior to field implementation

Revision history

Rev.	Date	Purpose	Description
00	15 Aug 2026	Draft for coordination	Initial English issue including the master methodology and method statement register

Document hierarchy

- Contract requirements, permits and statutory approvals.
- Approved project specifications and approved-for-construction (AFC) drawings.
- Approved design calculations, equipment schedules and coordinated builders-work drawings.
- Approved activity-specific method statements, risk assessments and Inspection and Test Plans (ITPs).
- Approved manufacturer installation, operation and maintenance instructions.
- This Master Construction Method Statement.

CONFLICT CONTROL

Work shall stop at the affected interface if drawings, BOQ descriptions, supplier instructions or site conditions conflict. The contractor shall raise an RFI and shall not resolve a dimensional, structural, hydraulic or electrical conflict through undocumented site judgement.

1. Purpose and Use

This document establishes the common construction controls for delivering the Hydrofarms NFT lettuce greenhouse project safely, in accordance with the approved design, and with a traceable quality record. It is intended for the client, consultant, general contractor, subcontractors, suppliers and commissioning team.

The document does not replace detailed activity methods. Its role is to define the mandatory project-wide sequence, interfaces, responsibilities, hold points, records and acceptance logic that every activity-specific submission shall follow.

Objectives

- Protect the approved geometry and performance requirements during construction.
- Prevent duplicated, omitted or out-of-sequence scope between civil, structural, hydraulic, cooling, electrical and automation packages.
- Ensure buried and concealed works are tested and accepted before covering.
- Create a direct link between the BOQ item, material approval, AFC drawing, method statement, ITP and inspection record.
- Deliver a greenhouse that is not only physically complete, but tested, commissioned, documented and ready for crop operation.

2. Approved Project Basis

Parameter	Approved basis for methodology
Production greenhouse	36 m wide x 126 m long; 14 bays at 9 m along the 126 m length
Service greenhouse	9 m x 36 m, located on the western side of the production greenhouse
Greenhouse levels	5.5 m underside / eaves reference and 7.0 m highest arch point
Production zoning	Eastern and western production circuits; central operating aisle 2.5 m
Table aisles	0.75 m clear operating aisles between production tables
Production table concept	16 m long table carrying 10 NFT channels; plant centres 200 mm in staggered pattern
Nursery zone	Six nursery tables, 17 channels per table, independent feed and gravity-return circuit
Nursery nutrient tank	3 m ³ , confirmed approved capacity
Production nutrient tanks	Two independent operating tanks serving eastern and western production zones
Reserve water	Four adjacent 10 m ³ tanks in one room and hydraulically interconnected as approved
Chillers	Five units at 7.5 TR each: two eastern, two western and one nursery
Exhaust fans	35 production fans and 3 service-greenhouse fans
Air circulation	28 production HAF fans and 2 service-greenhouse HAF fans
Cooling pads	150 mm pad depth with dedicated underground tank systems and distribution/collection assemblies
Covering	Solarwoven Ultra roof and applicable production sides; 10 mm local polycarbonate to the four service-greenhouse sides
Screens	Motorised external Harmony screen and motorised internal Harmony 3015 O E screen over production and service areas
Water path	Groundwater source to WTP, reserve storage, operating tanks, fertigation and independent pad-water systems

All dimensions shall be verified against the latest coordinated AFC drawings before setting out, procurement or fabrication. The values above establish the construction-method basis and do not authorise field changes.

3. Scope Boundaries and Interfaces

Included master scope

- Mobilisation, survey control, site logistics and temporary facilities.
- Earthworks, civil works, tank rooms, floors, interlock, drains and external connections.
- Greenhouse structural steel, bases, bracing, roof vents, rainwater system and protective treatment.
- Roof/side covering, polycarbonate, insect netting and motorised shading systems.
- NFT production and nursery tables, channels, supports, feed and return assemblies.
- Water treatment, reserve storage, nutrient tanks, pumps, filters, chillers and fertigation.
- Cooling pads, exhaust fans, circulation fans and associated supports and water systems.
- Electrical distribution, controls, sensors, germination lighting, earthing and emergency power interfaces.
- Testing, commissioning, training, first-cycle support and turnover records.

Interfaces requiring written confirmation

Interface	Mandatory coordination output
Well to WTP	Source flow, pressure, water analysis, incoming pipe, cable/control and ownership boundary
Structure to civil	Column/base coordinates, anchor templates, pedestal levels, grout and earthing bonds
Structure to covering/screens	Fixing rails, drive supports, clearances, deflection limits and weather seals
Civil to buried services	Sleeves, trenches, invert levels, chambers, pressure tests and as-built survey before covering
Tanks/pumps to electrical	Motor schedule, starting method, protection, isolators, cables, interlocks and dry-run protection
Cooling to structure	Independent fan/pad support frames, load paths, anti-vibration and service clearances
NFT to hydraulics	Table orientation, two-point feed, return slope, access and cleanability
Automation to all systems	I/O list, sensor locations, fail-safe states, alarm matrix and manual override
Packaging/germination to service greenhouse	Zone separation, hygiene flow, utilities, lighting and drainage

4. Project Organisation and Responsibilities

Role	Minimum responsibility under this method
Project manager	Overall delivery, programme, resources, commercial interfaces and approval status.
Construction manager	Daily sequence, work-front release, subcontractor coordination and protection of completed work.
Discipline engineer	Technical compliance, shop drawings, calculations, RFIs, setting-out data and method implementation.
QA/QC manager	ITPs, material inspections, calibration, inspection requests, NCR control and turnover dossier.
HSE manager	Risk assessments, inductions, permits, lifting plans, emergency response and site inspections.
Surveyor	Benchmarks, grids, levels, as-built coordinates and independent check before irreversible work.
Storekeeper	Approved-material identification, storage conditions, batch traceability and issue records.
Commissioning manager	Systemisation, pre-commissioning, functional testing, integrated trial and training.
Subcontractor supervisor	Briefing crews, task controls, inspection readiness, housekeeping and daily records.

5. Preconstruction Controls

Work-front release

No activity shall start until the responsible engineer signs a work-front release confirming all of the following:

- The latest AFC drawing and approved method statement are available at the work location.
- Materials and samples are approved, traceable and correctly stored.
- Required preceding inspections are accepted and no open NCR affects the work front.
- Setting-out points, access, lighting, lifting route and temporary utilities are ready.

- The crew has received the method statement briefing and task risk assessment.
- Inspection notification periods and consultant witness requirements are understood.

Submittals required before field work

- Coordinated shop drawings and builders-work drawings.
- Material approval requests, certificates, samples and supplier technical data.
- Method statement, task risk assessment and ITP.
- Survey setting-out package and benchmark confirmation.
- Lifting plan and temporary-works calculation where applicable.
- Equipment schedule, load schedule, valve schedule, I/O list and commissioning plan as applicable.

6. Master Construction Sequence

Stage	Work package	Main activities	Release evidence
01	Survey and investigations	Boundary verification, benchmark, grid, utilities scan, geotechnical confirmation and water analysis	Approved survey and investigation records
02	Mobilisation	Temporary storage, welfare, fencing, access, temporary power/water and HSE controls	Site establishment inspection
03	Earthworks and underground infrastructure	Excavation, tank rooms, sleeves, buried feed/return/drainage, earthing and tests	Concealed-work release
04	Foundations and floors	Footings, pedestals, anchors, interlock/concrete, drains and groundcover interfaces	Civil dimensional and strength acceptance
05	Structural erection	Columns, arches, longitudinal ties, bracing, vents, gutters and alignment	Structural completion certificate
06	Envelope and climate systems	Fans/pads, covering, polycarbonate, insect netting and motorised screens	Weather-tightness and movement tests
07	Process and MEP installation	Tanks, WTP, pumps, chillers, fertigation, tables, NFT channels, electrical and controls	Mechanical/electrical completion
08	Commissioning and handover	Cleaning, flushing, calibration, functional tests, integrated trial, training and as-builts	Turnover and acceptance dossier

SEQUENCE PROTECTION RULE

Buried services, grounding conductors, sleeves and drains shall be installed, surveyed and tested before floors or interlock are closed. Groundcover shall be installed after dirty overhead and pressure-testing work is complete and before table final placement where access requires it.

7. Site Logistics and Material Control

Logistics zoning

- Separate clean storage for NFT channels, fertigation components, sensors and covering materials.
- Dry, level storage for motors, gearboxes, pumps, electrical panels and chillers; factory openings kept sealed.
- Racked storage for pipes by diameter and material, with end caps and identification.
- Protected laydown area for galvanised steel; no direct contact with soil or standing water.
- Dedicated cutting/fabrication zone isolated from greenhouse coverings and crop-contact components.
- Defined lifting corridors that do not pass over installed tables, tanks or personnel.

Material receipt and traceability

The storekeeper and QA/QC inspector shall check purchase order, approved submittal, manufacturer, model/code, quantity, damage, certificates and storage requirements. Nonconforming or unapproved material shall be physically segregated and tagged.

Protection rules

Material/system	Mandatory protection
Galvanised steel	Use fabric slings; repair damaged zinc coating with approved zinc-rich repair system; prevent grinding contamination.
Greenhouse film/screens	Keep wrapped, shaded and off sharp surfaces; do not open until fixing rails and work front are clean.
Polycarbonate	Store flat and shaded; keep UV side identification; seal flute ends with approved tapes/profiles.
NFT channels	Store straight on continuous support; cap openings; prohibit walking or loading on channels.
Pumps/chillers/panels	Keep dry, upright and sealed; preserve desiccants/heaters where supplied; record insulation checks before energising.
Sensors/instruments	Store in original packaging; maintain calibration certificates and tag each instrument to its final I/O reference.

8. General Execution Methodology

8.1 Survey and setting out

- Establish at least two protected permanent benchmarks tied to the approved coordinate system.
- Set out greenhouse corners, structural grids, tank rooms, service greenhouse and external equipment zone.
- Check diagonals and cumulative bay dimensions independently before excavation and before anchor placement.
- Record coordinates and levels of buried services, sleeves and chamber inverts before covering.
- Issue an as-built survey after foundations, structural erection and final utility installation.

8.2 Civil and underground works

- Excavation shall follow approved side-slope or shoring requirements; groundwater and rainwater shall be controlled without weakening formation.
- Tank rooms may use the approved economical masonry/structural solution, but base slabs, retaining action, waterproofing and access loads shall follow the approved civil design.
- Buried water lines shall be laid on prepared bedding, with controlled bends, accessible valves and marker/identification provisions.
- No buried pressure or gravity line shall be covered before hydrostatic/leak testing, invert verification and as-built survey.
- Interlock and concrete floors shall maintain approved falls to drains and shall not bridge flexible joints or access covers.

8.3 Structural steel and greenhouse frame

- Verify foundation strength, anchor coordinates, projection and base elevation before column erection.
- Erect in stable sequences using temporary bracing; no incomplete frame shall be left unstable at the end of a shift.
- Install columns, arches, under-arch ties, longitudinal members and X-bracing to the approved colour-coded section schedule and connection details.
- Independent fan and cooling-pad support frames shall transfer loads to foundations and shall not rely on covering rails.
- Record bolt grade, tightening method, final alignment and galvanising repairs before envelope installation.
- Roof-vent frames, drives and rain gutters shall be trial-operated and water-tested before final covering closure.

8.4 Covering, polycarbonate, insect netting and screens

- Confirm clean fixing rails, smooth contact surfaces, approved fasteners and suitable weather conditions before pulling film or screens.
- Orient material according to manufacturer marks; maintain approved tension while allowing thermal movement.
- Install 10 mm polycarbonate to service-greenhouse sides with UV face outward, expansion clearance, end tapes and compatible profiles/sealants.
- Install insect netting continuously at the service corridor and door interfaces, extending to the approved high point without unsealed gaps.
- Install internal and external screen mechanisms with aligned shafts, supports, gearboxes, limit switches, torque protection and manual emergency operation.
- Complete full open/close cycles, alignment checks and obstruction tests before acceptance.

8.5 NFT tables and crop channels

- Fabricate production and nursery chassis using approved section schedules, jigs and corrosion protection; verify diagonals and support levels.
- Provide stable leg contact on the approved floor zone without creating drainage traps.
- Set production tables to 16 m channel length and ten NFT channels per table; nursery tables carry seventeen channels each.
- Position 50 mm planting openings at 200 mm centres in the approved staggered pattern, maintaining 50 mm / 100 mm first-hole offsets for odd/even channels and no opening outside the channel body.
- Maintain the approved longitudinal drainage slope, two feed points per production channel and unobstructed access to feed and return fittings.
- Flush all channels and protect crop-contact surfaces from welding, grinding, sealant and construction dust.

8.6 Water, nutrient and drainage systems

- Install the groundwater-to-WTP line and WTP-to-reserve storage line with verified ownership boundaries, isolation and sampling points.
- Install reserve tanks, transfer pumps and level controls before filling the operating nutrient tanks.
- Maintain independent eastern production, western production and nursery hydraulic circuits. The nursery operating tank is 3 m³.
- Use the approved 3-inch buried feed mains for production and nursery circuits, with pump suction/discharge sizes and branch routes as shown on AFC drawings.
- Install table feed manifolds below channel level, with individual valves and flexible feed hoses entering each channel at the approved two points.
- Install gravity returns with continuous fall, cleanouts/access points and no trapped high points, returning to the correct operating tank.
- Pressure-test feed lines; water-test gravity returns; flush and disinfect crop-water systems before commissioning.

8.7 Tanks, pumps and chillers

- Inspect tank rooms for dimensions, waterproofing, drainage, safe access and pump-removal route before tank placement.
- Install each tank on a level continuous base with independent access to fittings; interconnected reserve tanks shall be balanced as approved.
- Provide each pump with accessible isolation, non-return protection where required, strainer/filter, unions/flexible connectors, gauge points and dry-run protection.
- Install five 7.5 TR chillers at the approved southern external equipment zone with small dedicated weather shades and service clearance.
- Connect refrigerant circuits to approved SS316L immersed coils in the nutrient tanks, with isolation, control, insulation, condensate management and leak testing.
- Do not energise pumps or chillers until circuits are filled, vented, electrically tested and released by the commissioning team.

8.8 Cooling and ventilation

- Erect fan chassis to the ground-supported approved detail; check level, frame anchorage, fan centreline and maintenance clearance.
- Install 35 production exhaust fans and 3 service-greenhouse fans at approved common levels and sealing details.
- Install 28 production and 2 service HAF fans with consistent airflow direction, safe support and guarding.

- Install 150 mm cooling pads, 30 mm distributor media, treated-sheet frames, top distribution pipe and lower collection trough with uniform wetting.
- Provide five production pad tanks/pumps across four production sectors as approved and two service-greenhouse pad tank/pump systems; all plastic tanks are housed in the approved underground chambers.
- Test fan rotation, current, vibration, interlocks, airflow and pad wetting before climate performance testing.

8.9 Electrical, controls and fertigation

- Install the approved distribution hierarchy from utility/generator/ATS to MDB, MCCs/local panels and final equipment isolators.
- No cable shall be buried; route cables on approved exposed galvanised trays/supports with segregation of power and control circuits.
- Provide one identified protected feeder and local isolation point for every motor or equipment load, including all five chillers and all pump sets.
- Bond the greenhouse frame, equipment frames, tanks where applicable, trays and panels to the approved earthing system.
- Install sensors at representative locations away from direct pad spray, heat sources and short-circuit airflow; tag to the approved I/O list.
- Treat the fertigation package as an integrated supplier unit where approved; separately schedule only external support systems and interface connections.
- Calibrate EC, pH, temperature, level, pressure and flow instrumentation before wet commissioning.

9. Quality Assurance and Inspection

Inspection logic

Each activity-specific ITP shall identify the characteristic to be checked, applicable drawing/specification, acceptance criterion, responsible party, record, and intervention level. The following intervention codes shall be used:

Code	Meaning	Rule
H	Hold point	Work cannot proceed without written release by the designated approving party.
W	Witness point	Notice is issued; work may proceed after the agreed notice period if the witness does not attend.
S	Surveillance	QA/QC may inspect during routine work without stopping the activity.
R	Record review	Certificates, reports and checklists are reviewed and accepted.

Master hold and witness points

Point	Inspection stage	Type	Mandatory record
HP-01	Survey grid and benchmarks before excavation	H	Approved setting-out report
HP-02	Formation level / soil condition before blinding or base construction	H	Geotechnical/formation acceptance
HP-03	Reinforcement, formwork, sleeves and anchor templates before concrete	H	Pre-pour checklist
HP-04	Buried pressure and gravity lines before covering	H	Test report and as-built survey
HP-05	Tank-room waterproofing before protection/closure	W	Waterproofing inspection/test
HP-06	Foundation strength and anchor survey before structural erection	H	Concrete and survey release
HP-07	Structural alignment, bracing, bolts and coating repairs before covering	H	Structural completion record
HP-08	Roof-vent and rainwater tests before final envelope closure	W	Operation and water-test record
HP-09	Pressure/leak tests before pipe insulation or concealment	H	Hydrostatic/leak-test report
HP-10	Electrical insulation, continuity and earthing before energisation	H	Electrical test certificates
HP-11	Pump/chiller/fan pre-start inspection	H	Pre-commissioning checklist
HP-12	Water-treatment quality acceptance before nutrient-system filling	H	Laboratory/field water results
HP-13	NFT clean-water flow-uniformity test	W	Hydraulic balancing report
HP-14	Integrated climate and automation test	W	Functional performance report
HP-15	72-hour integrated trial and handover review	H	Signed acceptance dossier

Nonconformance and corrective action

- Identify and segregate nonconforming material or work immediately.
- Raise an NCR with factual evidence, affected drawings/items and containment action.
- Obtain approved disposition: repair, rework, use-as-is with engineering concession, or reject.
- Reinspect and close the NCR before dependent work is released.
- Capture recurring causes through corrective action and toolbox feedback.

10. HSE and Risk-Control Framework

Each activity method shall include a task-specific risk assessment. The following controls are mandatory project-wide and shall be expanded for the actual work front.

Hazard	Minimum control
Excavations / tank rooms	Permit, services scan, edge protection, safe access, atmosphere/water control, daily inspection and rescue planning where confined-space criteria apply.
Work at height	Approved access platform/MEWP, inspected harness system where required, tool tethering, exclusion zone and wind/weather limits.

Hazard	Minimum control
Lifting	Approved lifting plan, certified equipment, competent operator/rigger, tag lines, exclusion zone and verified load/centre of gravity.
Steel erection	Temporary stability plan, erection sequence, bracing before release and prohibition on leaving unstable frames.
Electrical	Lockout/tagout, dead testing, controlled energisation, RCD protection for temporary supplies and authorised personnel only.
Pressure testing	Calibrated gauges, defined test pressure, restrained ends, exclusion zone and controlled depressurisation.
Chemicals/fertilisers	SDS, compatible storage, bunding, ventilation, eyewash, PPE and controlled dosing/transfer.
Hot work	Permit, fire watch, extinguishers, removal/protection of films/screens and post-work inspection.
Heat stress	Work/rest regime, drinking water, shaded rest, acclimatisation and supervision during high-temperature periods.
Moving equipment	Guards, isolators, commissioning exclusion zones and test communication before fan/screen/vent movement.

Emergency readiness

- Post emergency contacts, assembly point and site access coordinates.
- Maintain first aid, fire extinguishers, spill kits and rescue equipment appropriate to active work.
- Conduct emergency drills for fire, electrical incident, excavation rescue and chemical exposure before the relevant high-risk activity.

11. Environmental and Biosecurity Controls

- Prevent cement washout, oils, refrigerant, chemicals and nutrient solution from entering soil or drainage uncontrolled.
- Route RO reject, washdown and nutrient purge only to approved collection/disposal routes, including the evaporation-pond system where applicable.
- Segregate metal, plastic-film, packaging, chemical and general wastes.
- Control dust before installing crop-contact systems, films, screens, sensors and electrical panels.
- After envelope closure, control access through the hygiene route; use clean footwear/tools and protect insect-net barriers.
- Complete final cleaning, flushing and disinfection before introducing seed, seedlings or nutrient solution.

12. Testing, Commissioning and Handover

Systemisation

The commissioning manager shall divide the project into systems and subsystems with defined boundaries, tag lists and completion certificates. Construction completion shall be confirmed before pre-commissioning starts.

Commissioning stage	Minimum activities	Release output
Construction completion	Visual inspection, punch listing, tags, access, guards, lubrication and documentation check	Mechanical/electrical completion certificate
Static testing	Pressure/leak, continuity, insulation, earth, tank water test, valve stroke and calibration	Static test pack
No-load functional test	Rotation, interlocks, limit switches, alarms, manual/auto modes and emergency stops	Functional test sheets
Wet commissioning	Fill, vent, flush, disinfect, balance, dose, sample and verify water quality	Wet commissioning report
Performance testing	NFT uniformity, chiller duty, pad wetting, airflow, temperature/RH response and WTP output	Performance report
Integrated trial	Automatic operation under representative load for at least 72 continuous hours	Integrated trial certificate
Handover	Training, O&M manuals, as-builts, spares, warranties, asset register and first-cycle support plan	Signed turnover dossier

Minimum performance evidence

- Treated-water quality and WTP production rate at the agreed feed-water condition.
- Flow and pressure at pump duty points and uniform distribution across representative remote NFT channels.
- Gravity return without overflow, air locks, standing water or cross-connection between operating tanks.
- Chilled nutrient-water temperature pull-down and stable control for eastern, western and nursery circuits.
- Cooling-pad wetting uniformity, fan rotation/current/vibration and greenhouse climate response.
- Screen and roof-vent full travel, limit operation, obstruction response and manual override.
- Alarm, fail-safe, generator/ATS and control-mode response in accordance with the approved cause-and-effect matrix.

Handover dossier

- Approved AFC and redline drawings, final as-built drawings and survey records.
- Material approvals, certificates, warranties and supplier contact list.
- ITPs, inspection requests, test reports, calibration certificates and closed NCRs.
- Equipment datasheets, O&M manuals, preventive-maintenance schedules and asset register.
- Training attendance, competency sign-off and first-cycle operational support plan.
- Initial spare parts, special tools, consumables and inventory handover.

13. Method Statement Coding and Control

The following code format shall be used to maintain direct traceability across disciplines:

DOCUMENT CODE FORMAT	
HF-GH01-MST-[DISC]-[NNN] HF = Hydrofarms GH01 = greenhouse project MST = method statement DISC = discipline code NNN = sequential number.	
Discipline code	Scope
GEN	General, survey, logistics and site management
CIV	Earthworks, concrete, masonry, floors, drainage and tank rooms
STR	Structural steel, greenhouse frame, vents and rainwater components
ENV	Covering, polycarbonate, insect netting and shading systems
NFT	Tables, channels, crop-contact fittings and nursery systems
PLB	Feed, return, drainage and water-transfer pipework

Discipline code	Scope
MEC	Tanks, pumps, chillers, WTP, cooling and ventilation
ELE	Power, lighting, earthing and emergency supply
CTL	Automation, instrumentation, fertigation control and integration
COM	Testing, commissioning, training and handover

Appendix A - Method Statement Register

Document code	Title	Discipline	Primary prerequisite	Release stage
HF-GH01-MST-GEN-001	Master Construction Method Statement	Master	This document	Rev. 00 issued
HF-GH01-MST-GEN-002	Survey, benchmarks and setting out	Survey	Approved layout / grid drawings	Required before mobilisation works
HF-GH01-MST-GEN-003	Site establishment, logistics and temporary facilities	General	Site logistics plan	Required before mobilisation
HF-GH01-MST-CIV-001	Excavation, dewatering, backfill and compaction	Civil	Geotechnical report / earthwork drawings	Required before excavation
HF-GH01-MST-CIV-002	Foundations, pedestals, anchors and grout	Civil/Structural	Foundation and anchor drawings	Required before concrete
HF-GH01-MST-CIV-003	Underground tank rooms, waterproofing and drainage	Civil/Mechanical	Tank-room coordinated drawings	Required before excavation
HF-GH01-MST-CIV-004	Buried sleeves, utilities and concealed pipework	Civil/Plumbing	Builders-work and pipe-route drawings	Required before floors
HF-GH01-MST-CIV-005	Interlock, concrete margins, floor falls and groundcover	Civil/Envelope	Flooring and drainage layout	Required before finishes
HF-GH01-MST-STR-001	Greenhouse structural-steel erection and alignment	Structural	AFC steel drawings / erection plan	Required before erection
HF-GH01-MST-STR-002	Bracing, bolts, connections and galvanising repair	Structural	Connection schedule / coating data	Required before enclosure
HF-GH01-MST-STR-003	Roof vents, drives and weather seals	Structural/Mechanical	Vent shop drawings	Required before roof closure
HF-GH01-MST-STR-004	Rain gutters, downpipes and outfall connections	Structural/Civil	Rainwater drawings	Required before envelope completion
HF-GH01-MST-ENV-001	Greenhouse film installation	Envelope	Film/fixing manufacturer details	Required before installation
HF-GH01-MST-ENV-002	Polycarbonate cladding and accessories	Envelope	Polycarbonate shop drawings	Required before installation
HF-GH01-MST-ENV-003	Insect netting and hygiene corridor closures	Envelope/Biosecurity	Netting and door details	Required before closure
HF-GH01-MST-ENV-004	Internal and external motorised shading screens	Envelope/Controls	Screen-drive coordinated drawings	Required before installation
HF-GH01-MST-NFT-001	Production table fabrication and installation	NFT	Approved table fabrication drawings	Required before fabrication
HF-GH01-MST-NFT-002	Nursery table fabrication and installation	NFT	Approved nursery-table drawings	Required before fabrication
HF-GH01-MST-NFT-003	NFT channels, planting holes and accessories	NFT	Channel and hole-pattern drawings	Required before drilling/installation
HF-GH01-MST-PLB-001	Buried feed mains and branches	Plumbing	Feed-network route and profile drawings	Required before trenching
HF-GH01-MST-PLB-002	Table feed manifolds, valves and two-point feeds	Plumbing/NFT	Table hydraulic detail	Required before installation
HF-GH01-MST-PLB-003	Table returns, buried gravity returns and drainage	Plumbing/NFT	Return profiles and invert schedule	Required before installation
HF-GH01-MST-MEC-001	WTP, reserve tanks and transfer systems	Mechanical	WTP/tank-room GA and P&ID	Required before equipment setting
HF-GH01-MST-MEC-002	Operating nutrient tanks, pumps and accessories	Mechanical	Tank/pump GA and connection schedule	Required before installation
HF-GH01-MST-MEC-003	Chillers, refrigerant lines and SS316L coils	Mechanical	Chiller/refrigerant shop drawings	Required before installation
HF-GH01-MST-MEC-004	Cooling pads, tanks, pumps and water distribution	Mechanical	Pad hydraulic and frame details	Required before installation

Document code	Title	Discipline	Primary prerequisite	Release stage
HF-GH01-MST-MEC-005	Exhaust fans, support frames and HAF fans	Mechanical/Structural	Fan frame and electrical details	Required before installation
HF-GH01-MST-CTL-001	Fertigation, stock solutions and EC/pH control	Controls/Process	P&ID, supplier package and I/O list	Required before installation
HF-GH01-MST-ELE-001	Electrical distribution, trays, cables and local isolation	Electrical	SLD, cable and load schedules	Required before installation
HF-GH01-MST-ELE-002	Earthing, bonding, testing and energisation	Electrical	Earthing layout and test plan	Required before energisation
HF-GH01-MST-ELE-003	Germination-zone lighting and service-area power	Electrical	Lighting and small-power layout	Required before installation
HF-GH01-MST-CTL-002	Automation, sensors, alarms and integrated controls	Controls	Control philosophy, I/O and cause/effect	Required before programming
HF-GH01-MST-COM-001	Cleaning, flushing and disinfection	Commissioning	Cleaning/flushing plan	Required before wet commissioning
HF-GH01-MST-COM-002	Pre-commissioning and functional testing	Commissioning	System test packs	Required before start-up
HF-GH01-MST-COM-003	Integrated performance trial and handover	Commissioning	Performance plan / handover index	Required before final acceptance

Appendix B - Required Activity-Method Attachments

- Activity-specific risk assessment and task briefing record.
- Inspection and Test Plan with hold/witness points.
- Approved material submittals and manufacturer instructions.
- Applicable AFC drawings and coordinated shop drawings.
- Work-front release checklist and permit requirements.
- Inspection checklists, test forms and calibration evidence.
- Emergency arrangements and lifting/temporary-work plans where applicable.
- Defined redline/as-built and turnover-record requirements.

Appendix C - Minimum Site Forms

Form code	Form title	Purpose
HF-GH01-FRM-001	Work-front release	Confirms prerequisites before activity starts
HF-GH01-FRM-002	Material receiving inspection	Links delivered material to approval and certificates
HF-GH01-FRM-003	Survey and setting-out record	Records coordinates, levels, diagonals and checks
HF-GH01-FRM-004	Inspection request	Requests consultant/client inspection under the ITP
HF-GH01-FRM-005	Concealed-work release	Authorises covering after test and as-built verification
HF-GH01-FRM-006	Pressure/leak test report	Records test medium, pressure, duration and result
HF-GH01-FRM-007	Electrical test certificate	Records continuity, insulation and earthing results
HF-GH01-FRM-008	Equipment pre-start checklist	Confirms mechanical/electrical readiness
HF-GH01-FRM-009	NCR and corrective action	Controls nonconforming work and closure
HF-GH01-FRM-010	Commissioning test sheet	Records functional and performance results
HF-GH01-FRM-011	Training and competency record	Documents operator training and acceptance
HF-GH01-FRM-012	Turnover dossier index	Tracks all handover deliverables

NEXT CONTROLLED ISSUE

Following approval of this master document and register, the first activity-specific submissions should be survey/setting out, excavation and underground works, foundations/anchors, and buried utilities. These documents control irreversible site work and therefore have the highest schedule priority.